Partial results on exact equilibria in concurrent reachability games

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Abstract

We prove exact subgame-perfect equilibrium existence in a finite two-player reachability game when, before either target is reached, one player has at most one decision state and at most two actions there. The opponent's actions and the transition graph are unrestricted. The proof uses a finite-policy perturbation and a return-time argument; deviations may be arbitrary behavioral functions of the public state history. We also compute the entire exact equilibrium payoff set of a three-state game. It is nonclosed, and its points have unbounded, exactly determined public-memory requirements. These are partial results for C73-24. Existence for every finite two-player concurrent reachability game remains unresolved here.

1 Question and scope

Let a finite stochastic game have two players, finite nonempty action sets, initial state s_0, transition law P, and targets F_1, F_2. At a public state history the players randomize independently. Their behavioral strategies use that history alone. A finite history is feasible if each of its transitions has positive probability under some joint action. For a feasible prefix h, let u_i^h be the probability that the full path, including h, visits F_i. Thus $u_i^h = 1$ when h already visits F_i. Exact subgame perfection means

$$u_i^h(\tau_i, \sigma_{-i}|_h) \le u_i^h(\sigma|_h) \quad \text{for every feasible } h,\ i \in \{1,2\},\ \text{and behavioral } \tau_i. \tag{1}$$

This includes prefixes having probability zero under σ; each continuation has its own probability law with the prefix fixed.

The general existence question is explicitly left open in [1, Section 1]; its Section 2 gives the history-sensitive conventions above. The principal partial result of this note is the following.

Theorem 1. *Suppose that, among states outside $F_1 \cup F_2$, player 1 has exactly one action except possibly at one state s_*, where it has at most two. Then an exact SPE exists. Its actions depend only on the current state and which targets have already been visited. There is no restriction on player 2's finite action sets or on the transition probabilities. The same assertion holds with the players interchanged.*

The restriction on player 1 is an additional hypothesis, not a property of general C73-24 instances. Section 4 gives a separate, fully classified example explaining why a limit of equilibrium payoffs need not itself be an equilibrium payoff.

2 Terminal-reward games and two elementary tools

Consider a finite set S of unresolved states and finitely many terminal outcomes. At s, actions a, b produce an unresolved transition kernel $K(s, a, b, t)$ with row sum at most one and, otherwise,

a terminal outcome with reward $r_i \in [0, M]$. Write $g_i(s, a, b)$ for its expected immediate terminal reward. The payoff is the expected terminal reward, with value zero if termination never occurs. The bound $M < \infty$ is fixed. Let $V_i^{x,y}$ denote the vector of actual payoffs under stationary behavioral policies x, y. A stationary profile that is a best response for each player from every state is an SPE, including after off-path prefixes. No action observations are used.

Lemma 2. *In a finite Markov decision process with bounded nonnegative terminal rewards and zero reward on nontermination, a pure stationary policy is optimal simultaneously from all states against all behavioral policies. A statewise convex combination with fixed global weights of stationary policies that are optimal from all states is also optimal from all states.*

Proof. For $0 < \lambda < 1$, discount a reward received on transition n by λ^n. The Bellman map is a contraction in the maximum norm; selecting a maximizing action at each state gives a pure stationary optimum against all policies. There are finitely many pure stationary policies. Along some sequence $\lambda \uparrow 1$, one policy π is therefore optimal at every term. For each behavioral competitor and each initial state, monotone convergence in its discounted optimality inequality proves that π is undiscounted-optimal.

Let v be the optimal value vector and let z be the stated convex combination. The Bellman equality $v = g_z + K_z v$ holds because it holds for every component policy. Any closed unresolved recurrent class C of z is closed under each component of positive weight: a nonnegative transition out of C cannot cancel in a convex combination. Each such component earns zero on C, so its optimal value v is zero there. Iterating the equality gives

$$v = \sum_{k=0}^{n-1} K_z^k g_z + K_z^n v.$$

In a finite chain, the probability of avoiding both termination and the closed unresolved recurrent classes tends to zero. Since $v = 0$ on those classes, $K_z^n v \to 0$. The sum converges to the actual terminal reward of z, proving optimality. □

Lemma 3 (Crossing stationary policies)**.** *Every finite two-player terminal-reward game as above has fully mixed stationary profiles (x_n, y_n) converging to (x_0, y_0) such that, for all sufficiently large n, x_0 is an all-state best response to y_n and y_0 is an all-state best response to x_n. If (x_0, y_0) terminates almost surely from every state, it is an exact SPE.*

Proof. Let Π_i be player i's finite set of pure stationary policies. A vector $w_i \in \Delta(\Pi_i)$ defines the stationary behavior

$$T_i(w_i)(a \mid s) = \sum_{\pi \in \Pi_i :\, \pi(s) = a} w_i(\pi).$$

The policy is not sampled once and remembered: the displayed distribution is used anew at every history ending at s. Fix $\mu(s) > 0$ with $\sum_s \mu(s) = 1$, and set

$$L_i(\pi, z) = \sum_s \mu(s) V_i^{\pi,z}(s).$$

For a fully mixed stationary opponent z, this is continuous in z. Indeed, the support graph with π fixed does not change while all of z's action probabilities stay positive. Its closed unresolved classes have value zero, and the other values are obtained from the inverse of the transient matrix $I - K$.

Put $N = \max_i |\Pi_i|$, fix $0 < \delta < 1$, and restrict each policy probability to be at least $\eta = \delta^N / N$. Against w_{-i}, prescribe the correspondence of all such w_i satisfying

$$w_i(\pi) \le \delta w_i(\pi') \quad \text{whenever } L_i(\pi, T_{-i}(w_{-i})) < L_i(\pi', T_{-i}(w_{-i})). \tag{2}$$

It has nonempty compact convex values. To see nonemptiness, order the $m = |\Pi_i|$ policies from best to worst and assign probabilities proportional to $1, \delta, \ldots, \delta^{m-1}$, breaking ties arbitrarily. The smallest is at least $\delta^{m-1}/m \geq \eta$. The graph is closed: every strict payoff inequality at a limit persists near that limit. Kakutani's theorem [3, p. 458] applies to the product of the two restricted simplices and gives a fixed point. This is a finite-policy version of the proper-pair construction in [2, Definition 2.3 and Theorem 2.4].

Take $\delta_n \downarrow 0$ and a convergent subsequence $w_i^n \to w_i^0$. If $w_i^0(\pi) > 0$, then eventually $w_i^n(\pi)$ is bounded below by a positive constant. Equation (2) rules out a strictly better pure policy, since it would force $w_i^n(\pi) \leq \delta_n$. Thus every policy in $\operatorname{supp}(w_i^0)$ maximizes L_i against $T_{-i}(w_{-i}^n)$ for all large n. By Lemma 2, one pure policy attains the optimum in every state. Positive weights $\mu(s)$ imply that any scalar maximizer also attains that optimum in every state. The convexity assertion of the same lemma proves the crossing property for $x_n = T_1(w_1^n)$ and $y_n = T_2(w_2^n)$.

If the limit profile terminates almost surely, $I - K_{x_0,y_0}$ is invertible, so compliance values are continuous in a neighborhood of it. For each fixed pure stationary deviation π, nonnegative rewards make its actual value lower semicontinuous in the opponent's policy: it is the increasing supremum of the finite-horizon values, which are continuous polynomials. Consequently, coordinatewise,

$$V_1^{\pi,y_0} \leq \liminf_n V_1^{\pi,y_n} \leq \lim_n V_1^{x_0,y_n} = V_1^{x_0,y_0}.$$

The argument for player 2 uses the other crossing inequality. Lemma 2 bounds every behavioral deviation by a pure stationary optimum. Hence neither player can gain after any feasible history. □

3 One binary decision state

Theorem 4. *In a finite two-player terminal-reward game with bounded nonnegative rewards, suppose player 1 has one action at every unresolved state except possibly s_*, where it has at most two. An exact stationary SPE exists against all behavioral public-state-history deviations.*

Proof. First solve the states where a player's reward is cooperatively unattainable. Let $H_i(s)$ be the maximum expected reward when a single controller chooses both players' actions, and let $Z_i = \{H_i = 0\}$. Lemma 2 supplies a pure stationary cooperative optimum. Nonnegativity implies that Z_i is closed under every unresolved transition of every joint action and that no positive i-reward can be earned there. On Z_1, use a joint stationary policy maximizing player 2's reward. Player 1 is indifferent and player 2 attains its cooperative upper bound, so this is an all-state SPE. Do the symmetric construction on Z_2. On their intersection both rewards are always zero; policies there may be made identical without changing any value. This solves the closed set $Z = Z_1 \cup Z_2$.

Replace first entry into $z \in Z$ by termination with the equilibrium reward just obtained. In every remaining state both cooperative values are strictly positive. Indeed, a finite path to a positive terminal reward either remains outside Z until termination or first reaches a state of Z where the relevant cooperative maximum was preserved. Against any fully mixed stationary opponent, either player therefore has a strictly positive optimal reward in every remaining state: it can prescribe its actions along such a finite positive-probability path. No bound uniform over fully mixed policies is needed.

If no state remains, we are done. If s_* has been removed or player 1 has only one action there, a pure stationary optimum for player 2 solves the remaining MDP. Otherwise apply Lemma 3. For large n, the crossed pair (x_n, y_0) terminates almost surely from every state. Player 2 is an exact best response to the fully mixed x_n, so its actual value is strictly positive everywhere. A closed unresolved recurrent class, whose actual reward is zero, is impossible.

If (x_0, y_0) is also terminating, Lemma 3 finishes the proof. Otherwise every closed unresolved recurrent class of this limit must contain s_*. A class avoiding s_* would remain closed under

(x_n, y_0), since player 1 has no choices within it. Thus there is just one such class C. Moreover, x_0 uses a single action A at s_*: if both actions had positive probability, C would again remain closed under the crossed pair. Call the other action B.

Fix y_0. Outside s_* the chain has no unresolved closed class avoiding s_*. Hence an excursion starting with either action at s_* reaches termination or returns to s_* almost surely. An A-excursion stays in C and returns almost surely without any terminal reward. For a B-excursion, let p be the probability of termination before return and b_i its expected terminal i-reward. Here $p > 0$: otherwise neither action could cause termination from s_*, contradicting termination under (x_n, y_0).

Any stationary policy using B with probability $t > 0$ on each visit terminates almost surely from s_*. Its value there is b_i/p, since

$$v_i = tb_i + (1 - tp)v_i.$$

An arbitrary player-1 deviation cannot exceed b_1/p there. To verify this without a finite-horizon assumption, enumerate its successive B-excursions and let q_k be the probability that its kth such excursion is started. Conditional on any starting history, that excursion has termination probability p and expected reward b_1. The terminating excursions are disjoint, so

$$\mathbb{E}[\text{terminal reward}] = b_1 \sum_{k \geq 1} q_k = \frac{b_1}{p} \Pr(\text{termination}) \leq \frac{b_1}{p}.$$

The identities follow by summing nonnegative quantities and also cover policies that eventually wait forever. Before the first visit to s_* or termination, a path from another state has no player-1 choice. Thus every stationary policy with $t > 0$ is an all-state best response to y_0. In particular x_n is one. Player 2 was already a best response to x_n, proving that (x_n, y_0) is an exact stationary SPE.

Finally restore the solved policies on Z. No deviation can return from Z to its complement, or improve its prescribed boundary reward. The remaining game's best-response inequalities therefore extend to the full game from every state and after every feasible prefix. □

Proof of Theorem 1. Record two public bits specifying which targets have been visited. If both bits are set, both players' rewards are fixed at one. If just player i's bit is set, that player is indifferent. Choose a joint pure stationary policy maximizing the other player's reachability probability. Lemma 2, applied with a success terminal reward of one, proves that this policy is an SPE in every such state. The policy can be implemented by independent pure actions and needs no information beyond the current state and the bits.

Before either target is reached, replace first entry into the solved region by its actual equilibrium payoff vector, of the form $(1, v)$, $(v, 1)$, or $(1, 1)$. This is a nonnegative terminal-reward game; paths never reaching either target give $(0, 0)$. It has the decision-state restriction of Theorem 4. Its stationary SPE, combined with the solved continuations, satisfies (1). The bits retain earlier target visits at every prefix, including off-path ones. □

4 A nonclosed exact equilibrium payoff set

Consider three states s, w, ℓ. States w and ℓ are absorbing, and both players' target is $\{w\}$. At s both choose Wait or Quit. Wait/Wait returns to s. Exactly one Quit goes to w or ℓ with equal probability; two Quits go to w surely. Infinite waiting loses.

Theorem 5. *The exact SPE payoff set from s, allowing arbitrary behavioral public-state-history strategies, is*

$$\{(u_m, u_m) : m \geq 1\}, \qquad u_m = \frac{m+1}{2m}. \tag{3}$$

For a deterministic public finite controller realizing u_m, at least m unresolved modes are necessary, and m suffice. The smallest mean termination time among all SPEs with payoff u_m is

$$\frac{(m+1)(2m+1)}{6m}. \tag{4}$$

Proof. Before termination a public history is specified by its length. Write p_n, q_n for the Quit probabilities at length n and u_n for the common actual continuation payoff in an SPE. Immediate Quit guarantees at least $1/2$, so $u_n \geq 1/2$ at every feasible prefix. Every unresolved tail has a first stage with a positive Quit probability; an all-Wait tail would instead have actual value zero.

In fact $u_n > 1/2$. If $u_n = 1/2$, move past any initial Wait/Wait stages to the first positive-quitting stage and write its probabilities as p, q and its next unresolved value as $v \geq 1/2$. The Bellman identity gives

$$\tfrac{1}{2} = pq + \tfrac{1}{2}[p(1-q) + (1-p)q] + (1-p)(1-q)v \geq \tfrac{1}{2} + \tfrac{1}{2}pq.$$

Thus exactly one of p, q is positive. The other player could Quit immediately and gain half that positive probability, a contradiction.

At any positive-quitting stage both probabilities must be positive: if one were zero, the other's supported Quit action would yield $1/2$, less than its value. A supported action must have the equilibrium one-step return, because changing just that action is an admissible deviation. If either probability is one, both are one: against certain Quit, Quit gives one and Wait gives $1/2$.

At a stage with $0 < p, q < 1$, indifference gives

$$u = \frac{1+q}{2} = \frac{1+p}{2}, \qquad \frac{1+p}{2} = \frac{p}{2} + (1-p)v.$$

Hence $p = q$, $v = 1/[2(1-p)]$, and $p \leq 1/2$ because $v \leq 1$. Wait/Wait stages preserve the value. If p' is the Quit probability at the next positive-quitting stage, then

$$p' = \frac{p}{1-p}, \qquad \frac{1}{p'} = \frac{1}{p} - 1. \tag{5}$$

This includes $p = 1/2$, followed by $p' = 1$. At any positive-quitting stage the reciprocal is either one or at least two. Repeated subtraction of one can stay in this set only if its initial value is an integer m. Otherwise it eventually lies strictly between one and two. Thus the positive Quit probabilities are forced to be $1/m, 1/(m-1), \ldots, 1$, separated only by finite Wait/Wait intervals. The initial payoff is necessarily u_m.

For attainment, use modes $k = m, m-1, \ldots, 1$. In mode k each player independently Quits with probability $1/k$. Every unresolved transition updates k to $\max(k-1, 1)$; this also defines behavior at off-path prefixes following mode one. For $k \geq 2$, the actual Bellman return is

$$\frac{1}{k} + \left(\frac{k-1}{k}\right)^2 u_{k-1} = u_k.$$

Quit gives $(1 + 1/k)/2 = u_k$, and Wait gives $1/(2k) + (k-1)u_{k-1}/k = u_k$. In mode one, Quit gives one and Wait gives $1/2$. Under any unilateral deviation, the other player still Quits surely by mode one. Termination therefore occurs within m stages, and backward induction proves optimality against every behavioral deviation after every feasible prefix.

A deterministic public finite controller assigns an action distribution to its current mode and physical state. Equation (5) forces m distinct positive Quit probabilities along the surviving histories of any SPE with payoff u_m; these require at least m modes. The construction attains the bound. In that construction survival after $k < m$ stages has probability $((m-k)/m)^2$. Summing these survival probabilities gives

$$\mathbb{E}\tau = \sum_{k=0}^{m-1} \left(\frac{m-k}{m}\right)^2 = \frac{(m+1)(2m+1)}{6m}.$$

Every other SPE with the same payoff can only insert Wait/Wait stages between the forced quitting stages. Such delays cannot reduce the mean termination time. This proves (4). □

In particular $(1/2, 1/2)$ is the limit of exact SPE payoffs but is not itself attained. The constructed policies converge pointwise, at each fixed unresolved history, to perpetual Wait/Wait, whose actual payoff is $(0, 0)$. This game nevertheless has the stationary SPE Quit/Quit. The memory lower bound concerns prescribed payoffs, not existence of some finite-memory equilibrium.

5 Remaining question

Theorem 4 does not cover a player with two decision states or three actions at its sole decision state. In these cases the single-delay reduction used in its proof is unavailable. Neither theorem settles the general existence or nonexistence question of C73-24.

Use of AI tools

The author used research AI agents that the author built and configured (GPT-6 Astra max, Claude Opus 5.5 max and GPT-6 Astra Pro) during the proof work, for proof search and construction, proof checking, translation, Python computation and similar tasks. The final manuscript was written, and the proofs were checked, by the human author. All responsibility for this work rests with the author personally.

References

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